Multi-cell Outdoor Channel State Information Dataset (MOCSID)

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Introduction

Motivation & Problem

- **Goal:** Build location-aware applications using CSI, without relying on GNSS.
- **Challenges with real-world CSI datasets:**
  - Hard and expensive to collect at large scale.
  - Single-cell setups.
  - Lack trajectory diversity and long-range mobility.
- **Our solution:** Use synthetic CSI datasets generated with ray-tracing simulations.

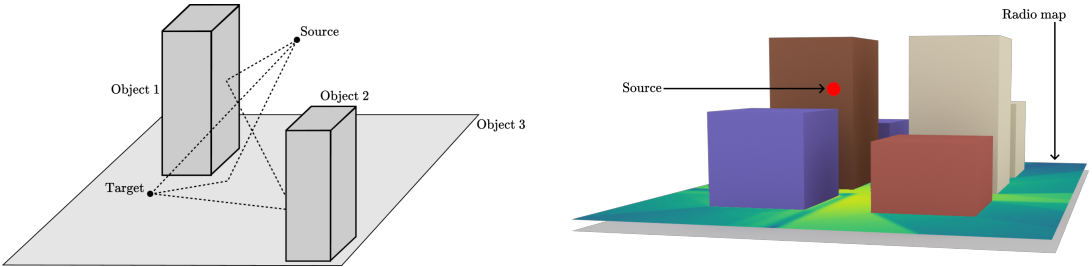
Examples of prior datasets:

- DICHASUS Euchner et al. 2021 – distributed CSI sounder for multi-AP setups
- MaMIMO-UAV Colpaert et al. 2023 – CSI dataset for aerial mobility

Why Use Synthetic Data?

- **Ray-tracing simulations** enable controlled and reproducible wireless environments
- Compared to real-world data:
  - Greater flexibility in modeling UE mobility and BS density
  - No need for costly measurements or position labeling

Let's now visualize what Sionna RT enables in practice.



Left: Illustration of ray tracing path solvers. Right: Illustration of coverage map. Source Aoudia et al. 2025

Key Requirements for the Channel Charting application

- **Spatial consistency:**
  - Nearby physical locations should produce Similar CSI
  - Requires realistic multipath and cell overlaps
- **Temporal continuity:**
  - CSI must evolve smoothly along the user trajectory
  - Enables reliable learning of user dynamics and movement patterns

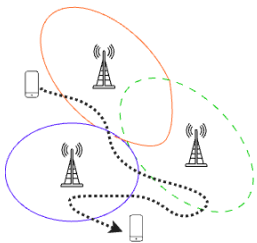


Illustration of spatial consistency through overlapping coverage regions.

Scenario Description

Scenario Description

- Scene data coming from OpenStreetMap
- **Used radio materials:**[1]
  - Concrete for buildings
  - Metal for rooftops
  - Medium Dry ground for terrain
- **Scene geometry:**
  - Scene area: $625\,\mathrm{m} \times 535\,\mathrm{m}$

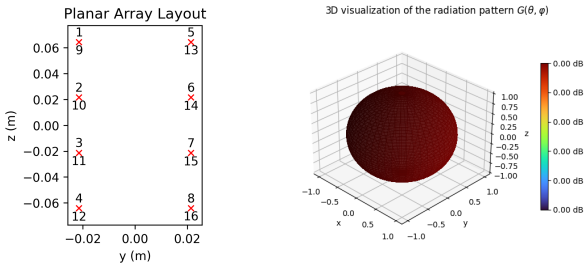


Scene layout with base stations (red spheres).

[1] 3GPP (Nov. 2020). *Study on channel model for frequencies from 0.5 to 100 GHz*. TS 38.901. Version 16.1.0

Base Stations Configuration

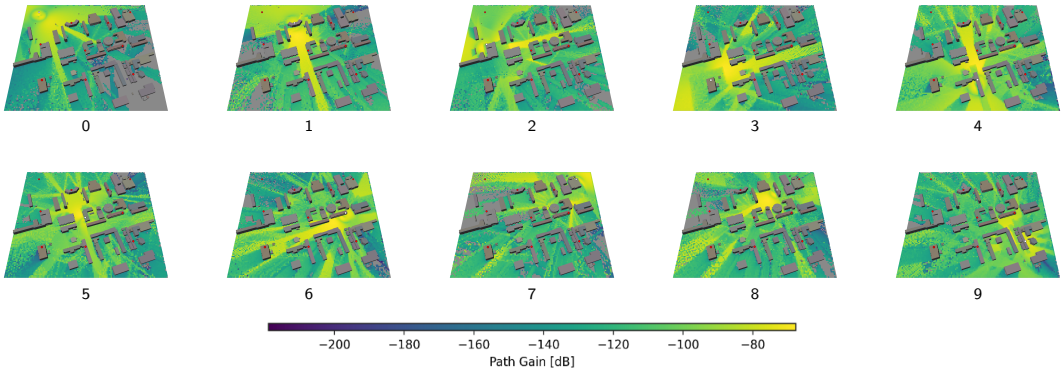
- **Deployment:** 10 base stations sharing the same array layout.
- **Array layout:** 4×2 rectangular antenna array (8 elements) with a Half-wavelength ($\lambda/2$) spacing.
- **Antenna elements:** Isotropic and dual-polarized ($\pm 45°$)



Left: 4×2 *antenna array layout. Right: Radiation pattern of an isotropic element.*

Path Loss Maps and Coverage Areas

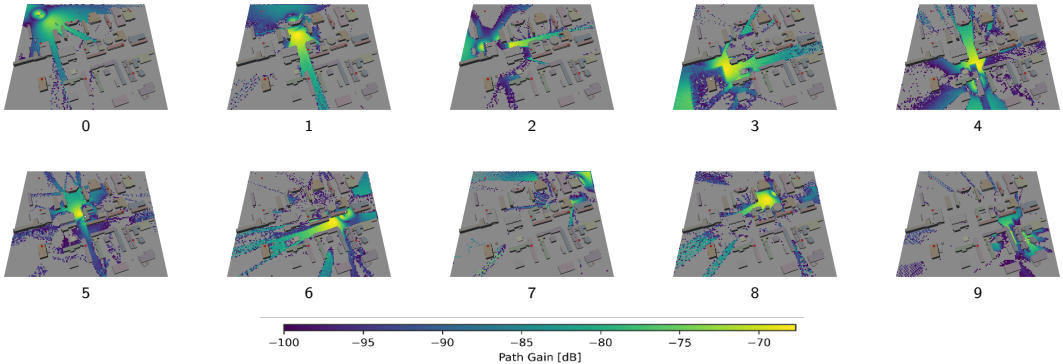
- Each image shows the path loss map at grid cells for one base station.



Path loss maps per base station.

Path Loss Maps and Coverage Areas

- Each image shows the path loss map at grid cells for one base station.

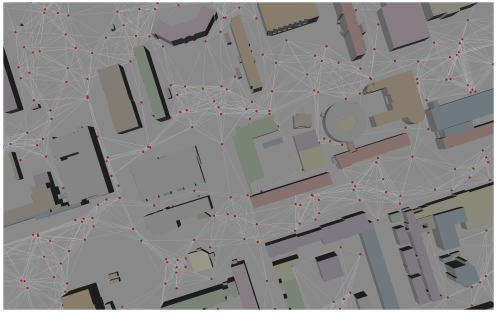


Thresholded path loss maps per base station.

Scenario Description – UE Mobility

Trajectory Generation using PRM [1]

1. **Graph Construction:** Sample obstacle-free points and connect nearby ones
2. **Path Planning:** Find shortest path from start to goal
3. **Smoothing:** Use cubic splines for realistic motion
4. **Sampling:** Resample at 10 Hz for continuous movement



Graph Construction

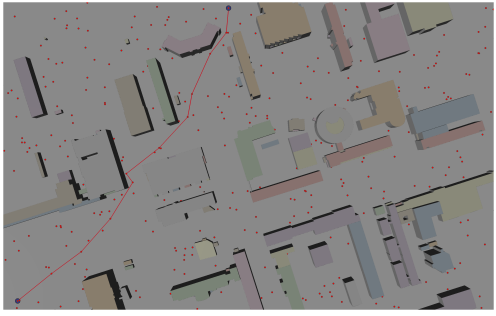
- **Antenna:** Single isotropic, vertically polarized
- **User Speed:** 1 ± 0.2 m/s

[1] Kavraki et al. (1996). Probabilistic roadmaps for path planning in high-dimensional configuration spaces.

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Path Planning

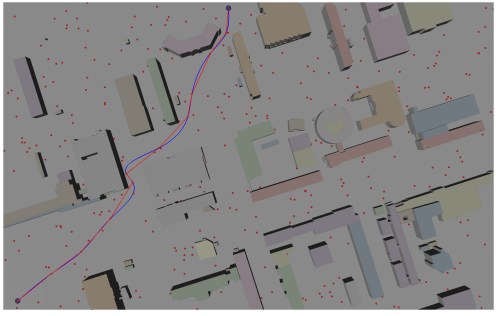
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Trajectory Smoothing

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- **User Speed:** 1 ± 0.2 m/s

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Scenario Description - Radio Parameters

Parameter	Value
Ray Tracing Setup	
Maximum depth of interaction	5
Number of sampled rays	10^6
Diffuse Scattering	Disabled
Synthetic array	Enabled
Radio Parameters	
Central frequency	3.5 GHz
Bandwidth	15.36 MHz
Subcarriers	256

Dataset Structure & Generation Costs

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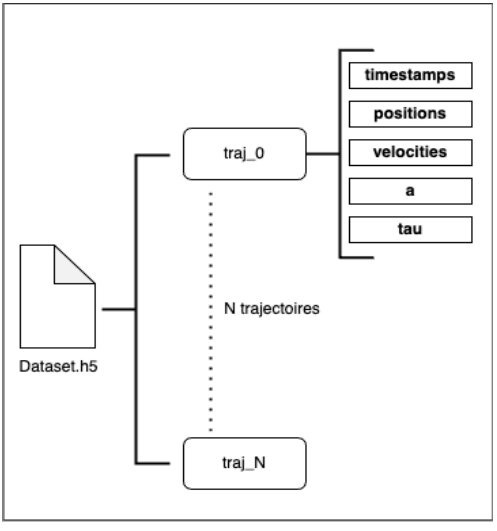
HDF5 Hierarchy

- Trajectory group per UE:
  - Timestamps
  - Positions
  - Velocities
  - Path coefficients
  - Path delays

Generation Costs

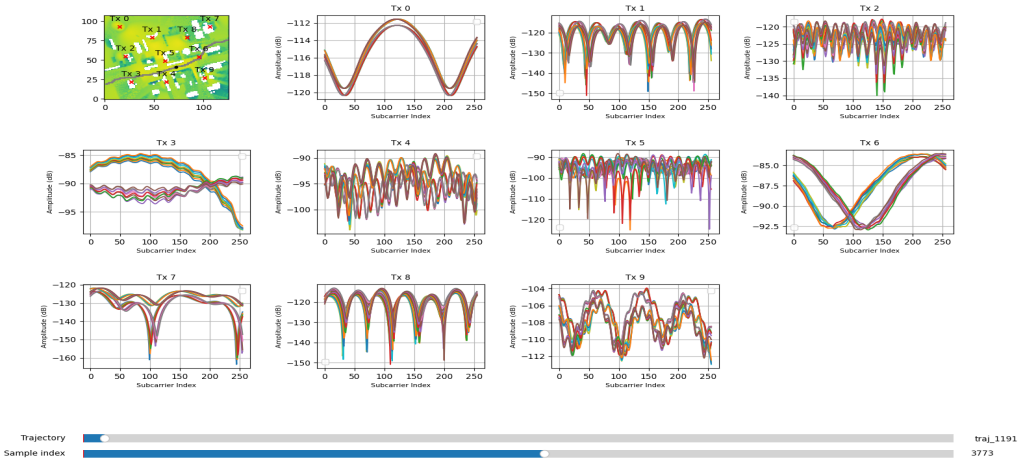
- Compressed sample size: 0.02 MB
- Generation time/sample: 34 ms (Using NVIDIA V100 GPU, 16GB)

  Should be faster on Sionna v0.1.0+



HDF5 structure visualization.

Dataset investigation and vizualizaiton

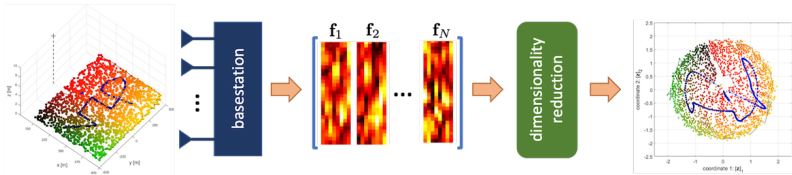


Amplitude per subcarrier per base station, visualization tools developed for the validation.

Channel Charting with MOCSID

What is Channel Charting?

- **Channel Charting (CC)** maps high-dimensional CSI into a low-dimensional **chart space**.
- The mapping is self-supervised — no labels or GNSS required.
- **Applications:**
  - UE pseudo-localization
  - Radio resources allocation
- **Challenge:** Geometry preservation from raw, noisy CSI



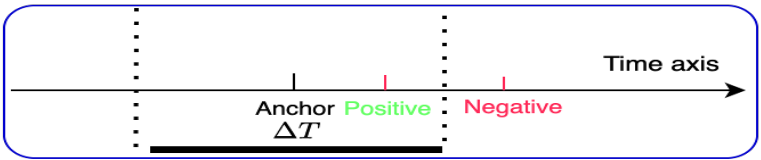
Example of Channel Charting pipeline. Adapted from Studer et al. 2018.

Channel Charting with MOCSID

- **Distance learning with contrastive learning -> Joint distance learning and dimensionality reduction** [1]
- **Triplet Margin Loss:**

$$\mathcal{L} = \sum \max\left(\|f_\theta(A) - f_\theta(P)\|^2 - \|f_\theta(A) - f_\theta(N)\|^2 + M,\ 0\right)$$

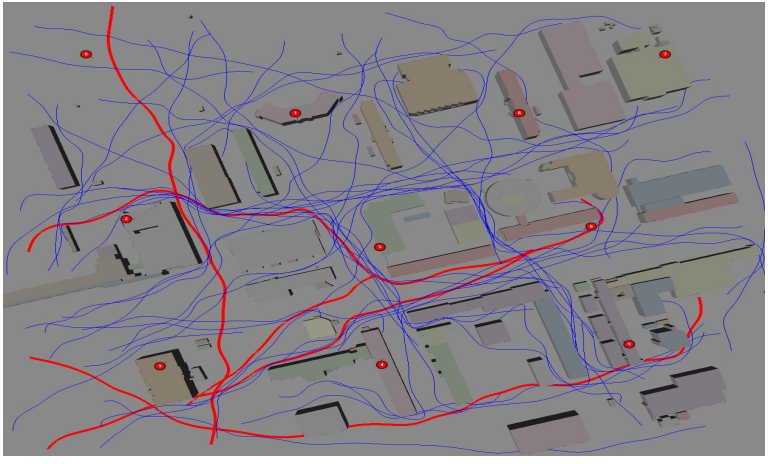
- Triplet selection based on short-term CSI correlation [2]



[1] Schroff et al. (2015), FaceNet: A unified embedding for face recognition and clustering.

[2] Ferrand et al. (2021), Triplet-Based Wireless Channel Charting: Architecture and Experiments.

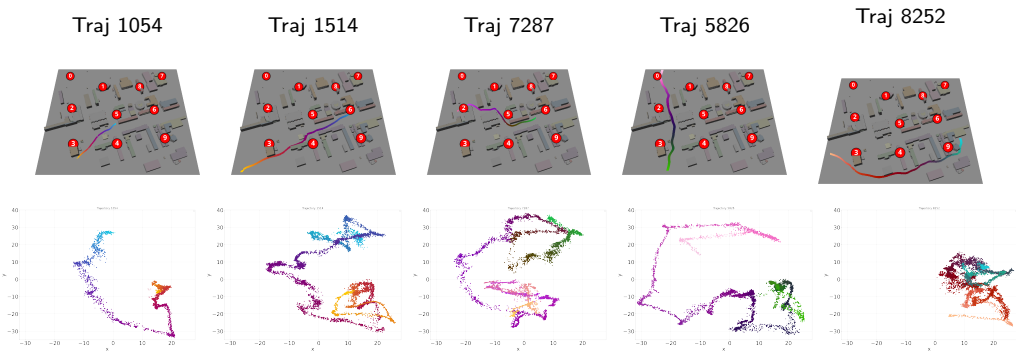
Train/Test Split



Training and test trajectories in real space (Blue = train, Red = test).

Trajectories lengths: $\approx$ 19 km (training), $\approx$ 2.5 km (test)

Charting Results: Real vs. Embedding



Top: Geometric locations coordinates — Bottom: Learned chart embeddings for 5 testing trajectories.

Charting Quality Metrics

- **Trustworthiness:** How well local neighborhoods are preserved (geometric location → chart)
- **Continuity:** Preservation in the reverse direction (chart → geometric location)
- **Kruskal Stress:** Global distortion of pairwise distances

Trajectory ID	Trustworthiness (%)	Continuity (%)	Kruskal Stress
1054	98.00	98.60	0.47
1514	97.47	99.17	0.43
7287	96.94	99.28	0.47
5826	98.27	99.10	0.49
8252	95.32	98.46	0.42
All	**92.00**	**98.96**	**0.46**

Conclusion

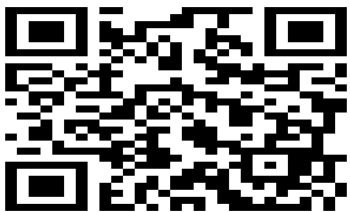
Conclusion & Dataset Availability

Key Features

- Multi-cell mobility within a realistic urban scenario with spatio-temporal consistency
- Can be extended or personalized
- Makhlouf, Guillaud, and Vindas 2025 - Multi-Cell Outdoor Channel State Information Dataset (MOCSID)

Availability

- Platform: Zenodo
- Size: 17 GB (compressed)
- Format: HDF5
- License: Open Database License



Questions about MOCSID: mohamed-el-mehdi.makhlouf@inria.fr

References I

- 3GPP (Nov. 2020). *Study on channel model for frequencies from 0.5 to 100 GHz*. TS 38.901. Version 16.1.0.
- Aoudia, Fayçal Aït et al. (2025). *Sionna RT: Technical Report*. arXiv: 2504.21719 [cs.IT]. URL: https://arxiv.org/abs/2504.21719.
- Colpaert, Achiel et al. (2023). *MaMIMO-UAV 3D Channel State Information Dataset*. Version V1. https://doi.org/10.48804/0IMQDF.
- Euchner, Florian et al. (2021). "A Distributed Massive MIMO Channel Sounder for "Big CSI Data"-driven Machine Learning". In: *International Workshop on Smart Antennas (WSA)*.
- Ferrand, Paul et al. (2021). "Triplet-Based Wireless Channel Charting: Architecture and Experiments". In: *IEEE Journ. Sel. Areas in Comm.* 39.8, pp. 2361–2373.

References II

- Makhlouf, Mohamed El Mehdi, Maxime Guillaud, and Yamil Vindas (June 2025). "Multi-Cell Outdoor Channel State Information Dataset (MOCSID)". In: *2025 Joint European Conference on Networks and Communications 6G Summit (EuCNC/6G Summit)*, pp. 85–90.
- Studer, Christoph et al. (2018). "Channel Charting: Locating Users Within the Radio Environment Using Channel State Information". In: *IEEE Access* 6, pp. 47682–47698. DOI: 10.1109/ACCESS.2018.2866979.
- Taner, Sueda, Victoria Palhares, and Christoph Studer (2024). *Channel Charting in Real-World Coordinates with Distributed MIMO*. arXiv: 2406.13722 [cs.IT]. URL: https://arxiv.org/abs/2406.13722.

Backup slides

Specular and diffuse reflections

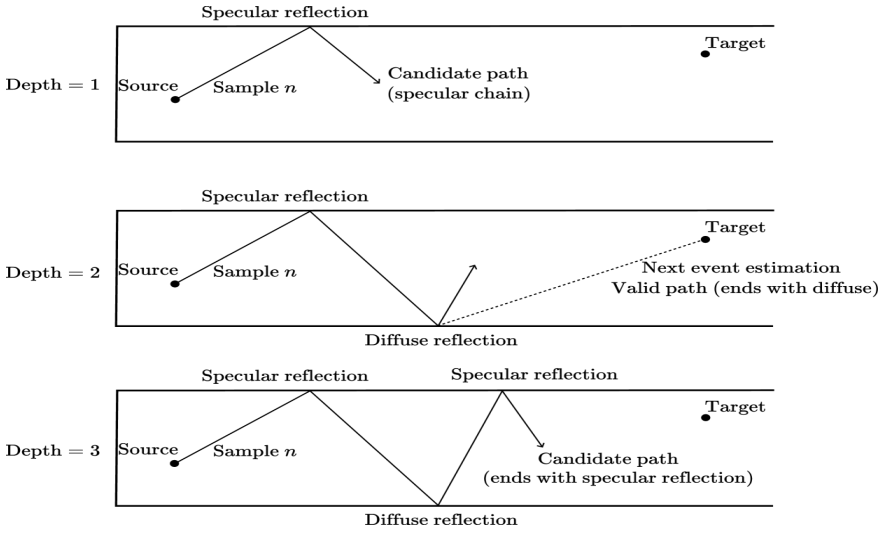


Illustration of specular and diffuse reflections with depth illustration

Model Architecture & Training Setup

Training Parameters

- Triplet selection with temporal window $T_c = 2\,\mathrm{s}$
- Triplet margin $M = 1$
- Optimizer: Adam
  - Learning rate: $\eta = 10^{-2}$
  - Weight decay: 10^{-7}
- Batch size: 1024 triplets
- 20 training epochs

Feature Extraction Taner, Palhares, and Studer 2024

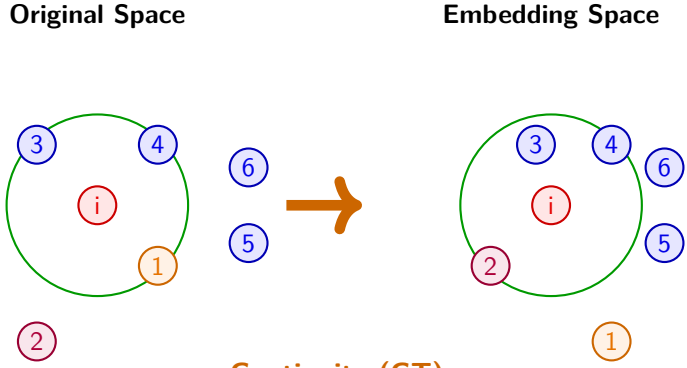
$$\mathbf{f}(n) = \frac{\left|\mathrm{vec}\left(\widehat{\mathbf{H}}(n)\right)\right|}{\left\|\mathrm{vec}\left(\widehat{\mathbf{H}}(n)\right)\right\|}$$

- Delay-domain CSI truncated to first $C \ll W$ taps
- Vectorization and magnitude normalization to remove phase and power variations

Network Architecture Ferrand et al. 2021

- Input: 1280-dimensional CSI feature vector
- MLP with Dense layers: 512, 256, 128, 64, 32 units
- Batch Normalization after each Dense layer
- Output: 2D embedding for channel charting

Illustrating Continuity

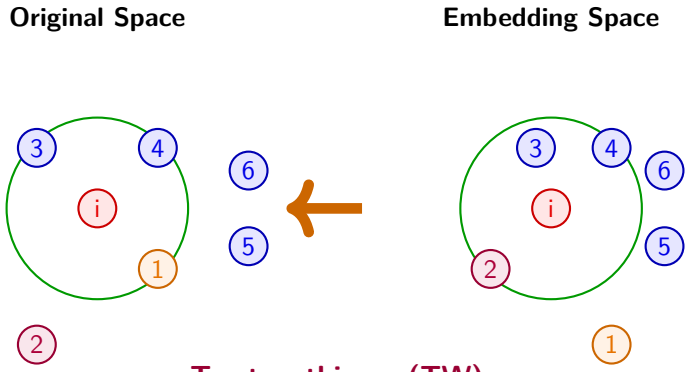


Continuity (CT)

- Measures if Original Space neighbors remain close in Embedding Space.
- **Penalty (False Negative)**: An original K-neighbor (e.g., pt 1) is *not* a K-neighbor in Embedding Space.

Guillaud et al 2024, Wireless Channel Charting for Next-Generation Radio Access Networks.

Illustrating Trustworthiness

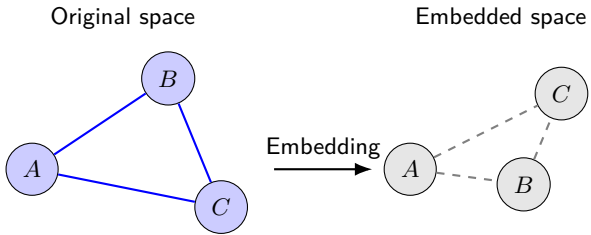


Trustworthiness (TW)

- Measures if Embedding Space neighbors were true neighbors in Original Space.
- **Penalty (False Positive)**: An embedding K-neighbor (e.g., pt 2) was *not* a K-neighbor in Original Space.

Both $CT, TW \in [0, 1]$; optimal if $CT = TW = 1$.

Kruskal Stress Illustration



Kruskal stress penalizes distortions in pairwise distances between original and embedded spaces.

Thank you for your attention

Questions?